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1. Microsoft Threat Modeling

Aim

To identify potential security threats and vulnerabilities in a given application scenario using Microsoft Threat Modeling Tool, and to recommend appropriate security controls.

Procedure

1. Install and open Microsoft Threat Modeling Tool.
2. Create a new threat model.
3. Define the application scenario and its components.
4. Identify external entities, processes, data stores, data flows and trust boundaries.
5. Draw the Data Flow Diagram (DFD) for the given scenario.
6. Define the properties of data flows and components.
7. Run the threat-generation/analysis feature.
8. Review the threats identified using the STRIDE methodology: Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege.
9. Analyze each identified threat and assign appropriate mitigation controls.
10. Generate/save the threat model report.

Output

Data Flow Diagram with trust boundaries

[Insert your screenshot here]

Result

The given application scenario was successfully modeled, potential STRIDE-based threats were identified, and suitable security mitigations were recommended.